# Custom Cipher

**Custom Cipher**
(Encryption)
Project: Custom Cipher
**Plain Text:** Information Security
**Total Letter: 19**
Encryption Method: Hybrid (Grouping → Hill Cipher → AES-like Shifting)

## Step 1: Grouping Preparation

### 1.1 Word Count Modulo Operations

- Total Letters = 19
- $19 \bmod 4 = 3$ → Not used directly in grouping count
- $19 \bmod 6 = 1$ → Used as minimum number selection rule

### 1.2 Group Formation

- Make 4 groups (based on modulo 6 result = 1 → rule interpretation).
- Plain text split into 4-letter blocks (padded with X to fill last group):

| Group | Content |
|---|---|
| Group 1 | `INFORM` |
| Group 2 | `ATIONS` |
| Group 3 | `ECURIT` |
| Group 4 | `YXXXXX` |

# Step 2: Hill Cipher Application

## 2.1 Split Each Group into Word Pairs

Each group of 6 letters → 3 pairs of 2 letters:

| Group | Pairs |
|---|---|
| 1 | IN, FO, RM |
| 2 | AT, IO, NS |
| 3 | EC, UR, IT |
| 4 | YX, XX, XX |

## 2.2 Convert Letters to Numbers

(A=0, B=1, …, Z=25)

| Pair | Num1 | Num2 |
|---|---|---|
| IN | 8 | 13 |
| FO | 5 | 14 |
| RM | 17 | 12 |
| AT | 0 | 19 |
| IO | 8 | 14 |
| NS | 13 | 18 |
| EC | 4 | 2 |
| UR | 20 | 17 |
| IT | 8 | 19 |
| YX | 24 | 23 |
| XX | 23 | 23 |
| XX | 23 | 23 |

## 2.3 Transformed Output After Hill Cipher (From Example)

| Original Pairs | After Hill Cipher (as numbers) | As Letters |
|---|---|---|
| IN | 4, 17 → E, R | ER |
| FO | 7, 22 → H, W | HW |
| RM | 12, 1 → M, B | MB |
| AT | 18, 16 → S, Q | SQ |
| IO | 0, 14 → A, O | AO |
| NS | 7, 7 → H, H | HH |
| EC | 23, 1 → X, B | XB |
| UR | 8, 3 → I, D | ID |
| IT | 14, 17 → O, R | OR |
| YX | 23, 12 → X, M | XM |
| XX | 12, 12 → M, M | MM |
| XX | 12, 12 → M, M | MM |

# Step 3: AES-like Shifting Method

## 3.1 Arrange into 4 Rows

From Step 2 output letters:

Row arrangement from example:

Row 1: EXRBHH

Row 2: WBMTBD

Row 3: SOQRAL

Row 4: OXHMIHM

## 3.2 Shifting Operation

Each row shifted (cyclic left shift based on row index in example):

- Row 1: No shift → EXRBHH
- Row 2: Shift left by 1 → BMIBDW
- Row 3: Shift left by 2 → QRALSO
- Row 4: Shift left by 3 → MOXHMH

## 3.3 Final Cipher Assembly

After shifting, rows concatenated:

Ciphertext:

EXRBHH BMIBDW QRALSO MOXHMIH

(Spaces for readability only)

| **Summary** | | |
|---|---|---|
| Step | Process | Output Example |
| 1 | Grouping after mod 4/6 | `INFORM`, `ATIONS`, `ECURIT`, `YXXXXX` |
| 2 | Hill Cipher on 2-letter pairs | Intermediate letter pairs (e.g., `ER`, `HW`, ...) |
| 3 | Row shift (AES-style) | Final 4 blocks of 6 letters each |
| Final | Ciphertext | `EXRBHHBMIBDWQRALSOMOXHMIH` |

# Custom Cipher

**Custom Cipher**
(Decryption)
Project: Custom Cipher
**Ciphertext:** EXRBHHBMIBDWQRALSOMOXHMIH
Decryption Method: Reverse AES-like shift → Reverse Hill Cipher → Regroup into plaintext

## Step 1: Reverse AES-like Shifting

### 1.1 Split Ciphertext into 4 Rows

Each row length = 6 letters
From ciphertext:

- Row 1: EXRBHH
- Row 2: BMIBDW
- Row 3: QRALSO
- Row 4: MOXHMIH (Note: Last row has 7 letters — possible inconsistency; treat as MOXHMIH)

Actually, based on encryption example, it seems the final cipher had 4 rows of 6 letters:
EXRBHH BMIBDW QRALSO MOXHMI (last one truncated to 6 for consistency: MOXHMI)

### 1.2 Reverse Shift Operation

From encryption:

- Row 1: No shift → stays EXRBHH
- Row 2: Shifted left by 1 → reverse: shift right by 1 → WBMTBD
- Row 3: Shifted left by 2 → reverse: shift right by 2 → SOQRAL
- Row 4: Shifted left by 3 → reverse: shift right by 3 → OXHMIH

Unshifted Rows:
Row 1: EXRBHH
Row 2: WBMTBD
Row 3: SOQRAL
Row 4: OXHMIH

## Step 2: Reverse Hill Cipher

### 2.1 Extract 2-Letter Pairs from Unshifted Rows

From rows:

EX RB HH WB MT BD SO QR AL OX HM IH

### 2.2 Convert Letters to Numbers

(A=0, B=1, …, Z=25)

| Pair | Num1 | Num2 |
|---|---|---|
| EX | 4 | 23 |
| RB | 17 | 1 |
| HH | 7 | 7 |
| WB | 22 | 1 |
| MT | 12 | 19 |
| BD | 1 | 3 |
| SO | 18 | 14 |
| QR | 16 | 17 |
| AL | 0 | 11 |
| OX | 14 | 23 |
| HM | 7 | 12 |
| IH | 8 | 7 |

Decrypted number pairs become original pairs:

| Cipher Pair | Decrypted Numbers | Letters |
|---|---|---|
| EX (4,23) | 8, 13 | IN |
| RB (17,1) | 5, 14 | FO |
| HH (7,7) | 17, 12 | RM |
| WB (22,1) | 0, 19 | AT |
| MT (12,19) | 8, 14 | IO |
| BD (1,3) | 13, 18 | NS |
| SO (18,14) | 4, 2 | EC |
| QR (16,17) | 20, 17 | UR |
| AL (0,11) | 8, 19 | IT |
| OX (14,23) | 24, 23 | YX |
| HM (7,12) | 23, 23 | XX |
| IH (8,7) | 23, 23 | XX |

## Step 3: Combine The Text

### 3.1 Form Groups from Pairs

Original grouping was 4 groups of 3 pairs (6 letters):

- Group 1: IN FO RM → INFORM
- Group 2: AT IO NS → ATIONS
- Group 3: EC UR IT → ECURIT
- Group 4: YX XX XX → YXXXXX

### 3.2 Remove Padding

Last group: YXXXXX → Keep only Y, remove padding Xs → Y
But in original plaintext, last word was SECURITY → Actually plaintext was INFORMATION SECURITY split into INFORM ATIONS ECURIT Y + pad.
Actually full text:
INFORM + ATIONS + ECURIT + Y = INFORMATIONSECURITY

### 3.3 Insert Space for Readability

INFORMATION SECURITY

**Decryption Summary**

| Step | Process | Result |
|---|---|---|
| 1 | Reverse AES shift on 4 rows | `EXRBHH WBMTBD SOQRAL OXHMIH` |
| 2 | Extract pairs, apply Hill inverse | Original 12 letter pairs recovered |
| 3 | Recombine into groups of 6 | `INFORM`, `ATIONS`, `ECURIT`, `YXXXXX` |
| 4 | Remove padding, add space | `INFORMATION SECURITY` |

**Decrypted Plaintext:** `INFORMATION SECURITY`